New challenges for an NSBI measurement of κ_λ in HH$\to b\bar{b}\tau\tau$, relative to the first NSBI measurement (ATLAS off-shell $H \to ZZ \to 4\ell$)

working note — draft for discussion

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Scope & one-line conclusion

This compares the systematics and background handling of the **first full NSBI measurement** (ATLAS off-shell $H \to ZZ \to 4\ell$; note HIGG-2023-01 = arXiv:2412.01600, Sandesara *et al.*) with what a κ_λ **measurement in HH**$\to b\bar{b}\tau\tau$ must do (baselined on CMS AN-18-121, Amendola *et al.*). The NSBI *skeleton* transfers directly; the new difficulty is almost entirely in the *inputs*: $b\bar{b}\tau\tau$ has a far messier final state, several **data-driven backgrounds with no per-event MC density** (QCD, jet$\to\tau$ fakes), an **irreducible same-final-state single-Higgs background**, an order of magnitude more (and larger) **experimental systematics that deform the very observable (m_{HH}) carrying the κ_λ signal**, and a **κ_λ-dependent theory uncertainty**. None of these are NSBI-specific — they hit any method — but several sit awkwardly inside a per-event density-ratio framework in ways the clean, all-MC off-shell analysis never faced. Page references to both notes are given inline.

Studies to do

Concrete studies to de-risk and execute this measurement, grouped by theme. Priority: **[P0]** blocking / do-first, **[P1]** important, **[P2]** nice-to-have; effort (S <1 wk, M weeks, L months). Trailing [§n] point to the relevant numbered section below. All abbreviations are defined in the **Glossary** at the end. Status tags: **[✓ done]** = completed and **[partial]** = partially completed in the preliminary phenomenological study (see the Progress box).

Progress — preliminary phenomenological result (Ghosh, Klute & Pan, 22 Jun 2026)

A first $b\bar{b}\tau\tau$ NSBI study (code in `NSBI-pheno/dihiggs_bbtautau`) has completed several items below. *The method and sensitivity are validated on **CMS full-simulation** samples (POWHEG NLO signal), but **without experimental-systematic nuisances in the fit**; a publishable phenomenological version additionally needs a DELPHES fast-sim reproduction (in progress — item #27, and the note's described setup). So the statistics and physics-method items are largely done, while the systematics, in-situ-CR, data-driven-background and Delphes-sample items remain open.* **Headline:** an unbinned 10-feature NSBI likelihood lifts the binned-m_{HH} secondary minimum near $\kappa_\lambda \sim 3$–4 and **narrows the 68% interval by ~an order of magnitude**, raising the $\kappa_\lambda = 0$ rejection from 1.1σ (binned m_{HH}) to $\mathbf{2.75\sigma}$ at $450\,\mathrm{fb}^{-1}$ and to $\mathbf{7.1\sigma}$ at the HL-LHC ($3000\,\mathrm{fb}^{-1}$, vs 2.8σ binned); 500-toy pulls $\sim \mathcal{N}(0,1)$ for both fits (Figs. 1–2, Table I). A focused test statistic (FTS) further tightens the 95% interval near the SM by 10–20% at no extra training cost (Fig. 3, Table II). **Completed: [✓ done]** #11, #14, #17. **Partial** (done without systematics, or by an alternative route): **[partial]** #1, #4, #7, #13, #15, #16, #18, #19, #20, #24, #25. Remaining items are systematics-, CR-, and data-driven-background-related.

Critical path

Statistics-first: port the JAX+IMINUIT fit (#11) and fix P_{ref} (#17), then build the in-situ-CR likelihood (#12) and the canonical ABCDnn flow with ExtendedABCD yields (#1,#2) in parallel; only *then* run Asimov closure (#13) and Neyman coverage (#14). The JES$\times\tau$ES/THU factorisation (#5), the coherent τES$\to$SVfit chain (#6), and the feature-ablation degeneracy study (#18) must land *before* coverage so the morphing is trustworthy; honest GP morphing-uncertainty (#9) is gated behind the MC-stat ensemble (#15). Multi-channel (#23), the k_{2V} tail (#21), POI-correlations (#22), and pipeline/pretraining (#25,#26) proceed as P1/P2 follow-ups.

Data-driven backgrounds

1. **[partial]** **[P0]** (L) **Canonical ABCDnn flow for QCD & jet$\to\tau$ fakes** (yield/shape split, three-piece uncertainty). MMD-trained NAF (continuous (N_j, N_b) conditioning, identity warm-up) morphs MC $\to$ the data-driven CR target; yield from ExtendedABCD/fake-factors, flow $\to$ per-event ratio $r_{\mathrm{QCD}}(x)$ via a flow-vs-reference classifier; port the three four-top nuisances as per-event morphing; sweep target dim 2D$\to$5D$\to$10D. **Done:** VR closure $|1-\langle f\rangle| < 0.05$, RMS < 0.15; $r(x) \geq 0$ integrating to the ABCD yield; max stable dimension documented. *(Prelim.: fake-τ and QCD are already given a per-event density via parametrised mis-ID* weights *on* POWHEG*/MG5 MC; the ABCDnn flow + closure on real data remains.)* [§4]

2. **[P0]** (M) **ExtendedABCD vs standard-ABCD yield closure for $b\bar{b}\tau\tau$ QCD.** Standard ABCD failed ∼18–19% at high N_j in the four-top analysis and the QCD-norm NP is already large; test the extended estimator's closure across 6–8 orthogonal SRs binned in (N_j, N_b). **Done:** bias $< 7\%$, RMS $< 12\%$ per multiplicity bin; a safe-vs-extended region map. [§4]

3. **[P1]** (S) **Encode the one-sided QCD shape NP into smooth morphing.** CMS uses one alternative template for both up/down; test mirrored / one-sided-exponential / half-Gaussian encodings so the poly-exp morphing of $r_{\mathrm{QCD}}(x)$ does not oscillate. **Done:** matches the CMS shape-impact rank ± 2, no oscillation, $< 2\%$ κ_λ bias. [§4]

4. **[partial]** **[P1]** (L) **Quantify the binned-QCD/fakes vs fully-unbinned cost.** Asimov (κ_λ= $0, 1, 5$): full unbinned vs MC-only NSBI with QCD/fakes as a fixed binned template; also bin the NSBI score at 5/10/20/50 to check convergence. **Done:** width inflation $< 15\%$, bias $< 0.3\sigma$, monotonic convergence; a bin-vs-unbin fallback decision tree. [§4, §7]

Systematics & learned morphing

5. **[P0]** (M) **Validate factorisation & cross-terms: JES$\times\tau$ES and THU$_{HH}\times\kappa_\lambda$.** Both JES and τES feed m_{HH} (a cross-term absent in clean 4ℓ) and THU$_{HH}$ is POI-coupled. Test $r_{\mathrm{comb}}/\prod_m r_m$ (median, KL) on held-out $\pm 1\sigma$ MC; compare naive-lnN vs κ_λ-coupled THU in Asimov. **Done:** factorisation residual 0.98–1.02, KL < 0.02; add a correction-NN if $> 2\%$. [§5]

6. **[P0]** (S) **Validate the coherent τES$\to$MET$\to$SVfit$\to m_{HH}$ recompute.** The weight-builder must shift τ energy at object level and recompute MET+SVfit+m_{HH} end-to-end, not shift m_{HH} directly. **Done:** matches the coherent truth to $< 2\%$ per-event MAE; the shortcut bias quantified. [§3, §5]

7. **[partial]** **[P1]** (M) **Validate poly-exp interpolation (κ_λ scan + $\pm 1\sigma$ anchors) under limited MC.** Intermediate κ_λ (0.5, 1.5) vs reweighted truth; downsample $\pm 1\sigma$ samples 100$\to$10%. **Done:** MAE $< 5\%$ down to ∼50% of MC, no gradient oscillation; minimum $\pm 1\sigma$ sample size to request from generators. [§1, §5]

8. **[P1]** (M) **Build & profile a CARL/DCTR per-event parton-shower/generator nuisance** (the NSBI upside). Train the per-event weight Pythia$\leftrightarrow$Herwig / ISR/FSR / VBF dipole-recoil; profile as one Gaussian NP vs the binned envelope. **Done:** KS $p > 0.1$, interval $\leq$ envelope with correct coverage, generator closure absorbed $< 2\sigma$. [§5]

9. **[P1]** (M) **Honest GP/ensemble-spread morphing-uncertainty with coverage calibration.** Fit a GP to per-event $r_{\text{true}} - r_{\text{pred}}$ residuals as a phase-space-dependent band (gated behind #15). **Done:** 65–70% coverage, calibration slope $1 \pm 10\%$, Spearman $\rho > 0.6$, narrower than a global band. [§5, §7]

10. **[P2]** (M) **Absorb b-tag/τ-ID SFs into the ratio, calibrated in situ** ($Z \to b\bar{b}$ / VZ($c\bar{c}$)). Demote the POG SF program to a cross-check and shrink the NN count. **Done:** in-situ SF within 30% of the POG prior, κ_λ not degraded, $|\Delta\text{pull}(0 \text{ vs } 5)| < 0.5\sigma$. [§5]

Statistics, fit & coverage

11. **[✓ done] [P0]** (M) **Port the jax+iminuit profile fit with the $\{\kappa_\lambda{}^2, \kappa_\lambda, 1\}$ basis** (root dependency). Replicate ATLAS Sec. 5: P_{ref}, the signal/interference/background mixture with κ_λ-dependent rate, $\pm 1\sigma$ ratio classifiers + poly-exp, profile fit with Gaussian-constrained NPs; cross-check vs a binned Asimov. **Done:** converges, Hesse invertible, parabolic NLL; unbinned-vs-binned width $< 5\%$. *(Prelim.: unbinned extended likelihood [Eq. 3] with cached density ratios + profile-NLL scan [Fig. 1]; binned m_{HH} cross-check via `iminuit`. The NP-profiling framework for the ∼100 experimental systematics is the remaining piece.)* [§1, §6]

12. **[P0]** (M) **Represent the in-situ control regions (18 DY SFs, $t\bar{t}$-SF, QCD B/D) inside the unbinned likelihood.** Add Poisson CR auxiliary terms sharing the same NPs; compare SR-only-free-NP vs SR+explicit CRs. **Done:** reproduce the CMS in-situ SFs within 10–20%, no double-counting, corr(SF,κ_λ)< 0.3. [§4, §6]

13. **[partial] [P0]** (M) **Asimov & closure: morphing bias under injected shifts and generator/Delphes mismatch.** Inject known $\pm 1\sigma$ on JES/τES/QCD/MC-stat and re-fit; fit data from chain X with a model trained on Y (Pythia↔Herwig; Delphes↔full-sim). **Done:** recover κ_λ to < 0.05, pulls $\sim N(0, 1)$, sim-closure bias < 0.1 (flag full-sim retraining if $> 10\%$). *(Prelim.: Asimov + 500-toy closure done on **full-sim**; the DELPHES↔full-sim closure is the open piece, paired with sample production [#27].)* [§5, §6]

14. **[✓ done] [P0]** (M) **Neyman construction & coverage validation on MC-truth toys.** 1000+ toys; build the belt at $\kappa_\lambda \in \{-1, 0, 1, 3, 5\}$. **Done:** empirical coverage $\geq 67\%$, $|\text{bias}| < 0.1$ at SM; any failure traced to a specific NP/component. *(Prelim.: 500 toys at $\kappa_\lambda = 1$ give pulls $\sim \mathcal{N}(0, 1)$ and calibrated 68% intervals for both NSBI and binned [Fig. 2]; extend to the full κ_λ grid and re-run with systematics.)* [§6]

15. **[partial] [P1]** (M) **MC-stat of the learned ratios: the wifi pilot + three-way estimator shootout.** Wrap the *existing* prelim ensembles as a frozen wifi basis (convex re-fit on a held-out split; sandwich covariance); propagate through the pointwise scan via Gong–Samaniego; *derive and validate the weighted-events extension* (NLO negative weights — not covered by the wifi paper); then the coverage shootout: ATLAS-style naive spread vs corrected bootstrap (fresh ensembles per pseudo-experiment) vs wifi, at nominal and $\pm 1\sigma$-downsampled stats (10/30/50%), $M \in \{8, 16, 32\}$. **Done:** wifi coverage 65–70% on $b\bar{b}\tau\tau$ toys incl. weighted events; the "training-stat = X% of $\sigma(\kappa_\lambda)$" decomposition produced; shootout verdict documented. *(See the item-6 update in §5: two independent strikes against the naive spread already exist.)* [§5, §6]

16. **[partial] [P1]** (S) **Interference-aware impact definition + binned-NSBI fallback.** Implicit-function-theorem impact at the minimum vs finite-difference and the CMS rank; histogram the per-event log-ratio (5/10/20 bins) as an exact, learning-free cross-check. **Done:** matches the CMS rank ± 2; binned constraint $\geq 80\%$ of unbinned, monotonic convergence. [§6, §7]

Physics & POI

17. **[✓ done] [P0]** (S) **Define & validate the reference hypothesis P_{ref}.** Compare SM / κ_λ-flat-mixture / pure-background / dedicated references; measure C2ST closure and how the $\{\kappa_\lambda{}^2, \kappa_\lambda, 1\}$ reconstruction degrades with $|\kappa_\lambda - \kappa_{\lambda\text{ref}}|$, including the thin high-m_{HH} k_{2V} tail. **Done:** C2ST $p > 0.05$

across the scan, $|\log r| < 10$, $< 5\%$ basis bias at extremes. *(Prelim.: a single* distant *reference $\kappa_\lambda = 5$ was chosen; a pooled-mixture reference gave 0.50 classifier accuracy [target–reference self-overlap] and was rejected; $r_{\text{ref}} \equiv 1$; stable against reference-sample replacement.)* [§1]

18. **[partial] [P0]** (M) **Feature-ablation degeneracy-breaking: $\rho(\kappa_\lambda$,JES/τES) and VIF vs feature schedule.** $m_{HH} \to$10-feature$\to$ +constituents$\to$ +SVfit covariance$\to$ +resolution handles; ablate to find the critical features and the asymptotic ρ floor. **Done:** reproduce $\rho\,1.000 \to 0.646$; a minimal set with $\rho < 0.7$, VIF< 1.5; settle whether constituents/SVfit push $\rho < 0.60$ or it plateaus. *(Prelim.: the* degeneracy-break *is demonstrated — the 10-feature NSBI lifts the binned-m_{HH} secondary minimum near $\kappa_\lambda \sim 3$–4 [Fig. 1]; features selected by total-variation distance + integral-closure, stable under one-at-a-time low-TV removal. The $\rho(\kappa_\lambda$,JES/τES)/VIF quantification needs the systematic variations.)* [§2, §5]

19. **[partial] [P1]** (S) **Source the κ_λ-differential THU$_{HH}$ band — and measure the shape-truncation bias.** Compile the scale/PDF/m_t band vs $\kappa_\lambda \in [-2, 5]$ and parametrise it as a κ_λ-dependent *rate* weight; then fit Asimov data generated with scale-varied *shapes* (per-event μ_R/μ_F weights, no new samples) with the normalisation-only model in *both* the binned and NSBI fits — two-sided: is the truncation *differentially* unsafe for the unbinned fit? **Done:** a smooth band reproducing the AN-25-103 Table-40 values (incl. the m_t sign flip); truncation bias quantified for *both* fits (< 0.05 each $\Rightarrow$ CMS-parity default validated; NSBI $\gg$ binned $\Rightarrow$ C4b required for our method). *(Prelim.: the total-level band per κ_λ now exists — AN-25-103 Table 40: $+4/-18\%$ at κ_λ=1, $+4/-22\%$ at 2.45, $+13/-4\%$ at 5 [m_t scheme]; the* component-differential *split needed for shape morphing remains open — C4b, see §5.)* [§2, §5]

20. **[partial] [P1]** (M) **Validate the LO$\to$NLO κ_λ-reweighting that generates the $R(x|\kappa_\lambda)$ sources.** Direct-NLO vs 2D-LO-reweight vs dedicated-NLO-reweight; propagate the difference through an Asimov fit. **Done:** NLO-reweight reproduces direct to $< 5\%$/bin and $< 0.1\sigma$ κ_λ bias; the LO-only cost quantified. *(Prelim.: sidestepped — signal generated* directly *at NLO [POWHEG] at $\kappa_\lambda \in \{0, 1, 5\}$ with a 3-point quadratic basis [the $\kappa_\lambda = 2.45$ point was dropped for morphing stability]; this study still applies if the CMS LO+reweighting samples are used instead.)* [§2]

21. **[P1]** (M) **Measure k_{2V} sensitivity in the high-m_{HH} VBF tail via unbinned NSBI.** VBF-enriched ($M_{jj} > 400\,\text{GeV}$), k_{2V}-morphed MC; Asimov-fit binned vs unbinned vs hybrid; cross-validate the tail (train low-m_{HH}, test high). **Done:** NSBI k_{2V} error 20–40% smaller, tail-extrapolation bias $< 10\%$, a minimum viable bin granularity. [§2]

22. **[P1]** (L) **Map the rate-broken POI correlations ($\kappa_t \leftrightarrow \kappa_\lambda$, $k_V \leftrightarrow k_{2V}$) and confirm the scope caveat.** Multi-POI grid; profile-likelihood correlation contours; attribute information to shape vs rate. **Done:** a POI correlation matrix; NSBI lowers $\rho(\kappa_\lambda,k_{2V})$ but does *not* improve the rate-broken pairs. [§2, §7]

23. **[P1]** (M) **Build the multi-channel + boosted simultaneous unbinned likelihood.** Per-channel vs joint morphing for the three $\tau\tau$ channels + boosted, with the CMS year/channel NP-correlation scheme. **Done:** combined κ_λ width within 5% of the best combination; no extra $> 0.5\sigma$ pulls. [§1, §3]

ML, samples & pipeline

24. **[partial] [P1]** (M) **Low-signal robustness: morphing calibration & uncertainty honesty at $\mathcal{O}$(10–100) signal events.** Downsample the $\pm1\sigma$ sets and effective signal yield (50/100/250) and retrain per level. **Done:** C2ST $p > 0.05$ at the $\pm1\sigma$ floor, ensemble spread $< 2\times$ the high-stat value; flag the binned fallback if it fails. [§5, §7]

25. **[partial] [P2]** (L) **Reproducible Perlmutter pipeline for the ∼2000-NN morphing ensemble.** Containerise Sophon+JAX+IMINUIT, a training manifest, resume-on-failure; benchmark per-NP classifiers vs a single conditional morphing network. **Done:** 500 NNs in $< 4\,\text{h}$ (∼1 GPU-day), $> 99\%$ resume, bit-identical reruns; the per-NP-vs-conditional trade-off documented. [general]

26. [**P2**] (M) **Pretraining follow-up: signal-vs-$t\bar{t}$ foundation-model benchmark vs scratch** (confirmatory). Scratch vs $t\bar{t}$-kinematics-pretrain vs HH-kinematics-pretrain; F-score $\kappa_\lambda = 0$ vs 5 by m_{HH} region. **Done:** signal-vs-$t\bar{t}$ pretraining closes $\geq 30\%$ of the FM deficit, else confirm scratch remains the baseline. [established null]

27. [partial] [**P1**] (M) **Produce & validate delphes $b\bar{b}\tau\tau$ samples for the phenomenological-paper version.** The sensitivity is currently validated on **CMS full-simulation** samples; a publishable pheno paper needs a DELPHES fast-sim reproduction. Validate the tuned CMS DELPHES card (b-tag, τ-ID/DeepTau, MET, lepton response) against the full-sim object performance and the reconstructed-vs-generated m_{HH} response, then regenerate signal + backgrounds and re-fit. **Done:** DELPHES-vs-full-sim closure on key observables (especially m_{HH} below 400 GeV) within the per-event-weight tolerance; the κ_λ interval reproduced to $< 0.1\sigma$. *(Prelim.: a standalone DELPHES card-validation pipeline is built; full sample production for the pheno paper is in progress. Pairs with #13.)* [§3, general]

1 The two measurements and the shared NSBI skeleton

ATLAS off-shell [ATLAS p. 1,5]: measures the off-shell signal strength $\mu_{\text{off-shell}}$ (hence the Higgs width) in $gg \to (H^* \to)ZZ \to 4\ell$, using $139\,\text{fb}^{-1}$. The physics is a *coupling that morphs an interference shape* (triangle–box in $ggZZ$). **CMS HH→ $b\bar{b}\tau\tau$** [CMS p. 3,4]: searches for non-resonant HH via ggF+VBF in $b\bar{b}\tau\tau$ ($\mathcal{B} = 7.3\%$), $137.6\,\text{fb}^{-1}$, three $\tau\tau$ channels ($\tau_e\tau_h, \tau_\mu\tau_h, \tau_h\tau_h$, plus boosted), extracting κ_λ (and k_{2V}, k_V). $\sigma^{HH}_{ggF} = 31.05^{+2.2}_{-5.0}\%$ fb, $\sigma^{HH}_{VBF} = 1.726$ fb [CMS p. 4].
What transfers (the skeleton). The ATLAS recipe (Sec. 5) is reusable wholesale: a reference-hypothesis P_{ref} so all NNs estimate ratios to a fixed point; a search-oriented *mixture model* for signal/interference/background; nuisance parameters in both the rate and the per-event density; a profile-likelihood test statistic in a custom JAX+IMINUIT fit; and an interference-aware impact definition [ATLAS p. 89,90]. The off-shell interference structure $(\mu - \sqrt{\mu})\,S + \sqrt{\mu}\,\text{SBI} + (1 - \sqrt{\mu})\,B$ [ATLAS p. 81] maps onto the κ_λ morphing basis $\{\kappa_\lambda^2, \kappa_\lambda, 1\}$ (since $gg \to HH = \kappa_\lambda \cdot \text{triangle} + \text{box}$). *The challenges below are about the inputs to this skeleton, not the skeleton itself.*

2 Physics / parameter-of-interest challenges

- **The POI lives in a shape, and that shape is m_{HH}.** κ_λ sensitivity is concentrated in the m_{HH} spectrum (triangle–box interference), strongest in the resolved regime $m_{HH} \in [250, 400]$ GeV. CMS encodes the κ_λ dependence by *reweighting* the signal MC — a 2D $(m_{HH}, |\cos\theta^*|)$ LO reweighting and a dedicated NLO reweighting for the final extraction [CMS p. 6]. In NSBI this becomes the $R(x|\kappa_\lambda)$ factor.
- **Multiple correlated POIs / EFT directions.** Beyond κ_λ the analysis also probes k_{2V}, k_V and HEFT operators (c_2, c_{2g}, c_g) [CMS p. 4]; the off-shell measurement is effectively one POI. More POIs $\Rightarrow$ a higher-dimensional morphing.
- **k_{2V} sensitivity lives in the high-m_{HH} VBF tail — where binning hurts most.** The $HHVV$ contact term grows with energy unless $k_{2V} = 1$, so VBF-HH k_{2V} sensitivity is concentrated in the high-m_{HH} tail; current analyses use only ∼4–5 bins there. This is precisely the regime where coarse binning discards the most information, so the unbinned NSBI gain is largest on the k_{2V} (and the low-m_{HH} κ_λ) shape directions. (k_{2V} is now the best-measured "inaccessible" coupling — the 2026 ATLAS+CMS combination gives $k_{2V} \in [0.73, 1.3]$.)
- **Interference impact ambiguity (shared, but worse).** Both have interference, so the finite-difference NP-impact definition is ambiguous; ATLAS replaces it with a local implicit-function-theorem impact [ATLAS p. 90]. κ_λ needs the same fix.

3 Final state and reconstruction

The off-shell 4ℓ final state is *exquisitely clean*: four leptons, fully reconstructed, with tiny energy-scale systematics and no jets/MET/τ/b-tagging in the core. $b\bar{b}\tau\tau$ is the opposite:

- **b-jets** (JES/JER, b-tagging) + **hadronic** τ (τ-ID, τ energy scale) + **MET** (the τ neutrinos). The $\tau\tau$ mass is *not* fully reconstructed — it requires an algorithm (SVfit); shifting τ energy shifts MET *and* SVfit coherently [CMS p. 93].
- **Three channels** with different background compositions and systematics, plus a boosted regime.

Consequence: the observable x for $b\bar{b}\tau\tau$ is softer and carries many more systematic "handles" than 4ℓ.

4 Background handling — the largest qualitative difference

The crux

Off-shell backgrounds are **all MC-based and clean** (one dominant irreducible $q\bar{q}ZZ$ continuum, modelled in simulation, plus EW). $b\bar{b}\tau\tau$ backgrounds are a **zoo, several of them data-driven**, and *data-driven backgrounds have no per-event MC probability density* — which is the object NSBI needs. This is the single hardest conceptual mismatch between $b\bar{b}\tau\tau$ and the off-shell NSBI template.

ATLAS off-shell (Sec. 2.5, Sec. 5). The dominant background is the irreducible $q\bar{q} \to ZZ \to 4\ell$ continuum, which *interferes* with the signal and is carried in the mixture model as a component $\mu_{q\bar{q}ZZ}$ (three floating NPs) constrained *in situ* by a dedicated n_{jets} discriminant in a control region [ATLAS p. 81]. Plus a small EW component. Everything is MC, so every process has a per-event density the NSBI ratios can target.

CMS $b\bar{b}\tau\tau$ (Sec. 7). A heterogeneous set, with the two largest estimated *from data*:

- **QCD multijet — data-driven ABCD** [CMS p. 61,62]. Four regions in (charge $\times$ τ-isolation); shape from region C (data minus other-MC), yield from $C \times B/D$. The QCD *normalisation* uncertainty ranges **5–100%** by category (Table 30, [CMS p. 92]); its *shape* uncertainty (alternative B-vs-C templates, symmetrised) is **one of the highest-ranking nuisances** in the most sensitive $\tau_h\tau_h$ channel [CMS p. 91,96,97].
- **Drell–Yan (Z+jets) — MC normalised by 18 data-driven scale factors** [CMS p. 63]. A simultaneous fit of $M(\mu\mu)$ across **18 control regions** (3 b-jet multiplicities $\times$ 6 Z-p_{T} bins), giving 18 normalisation SFs ($\sim$0.6–1.7) *and* 18 shape templates (scaled $\pm$) propagated with their covariance [CMS p. 65,94].
- **$t\bar{t}$ — MC shape + data-driven normalisation SF** [CMS p. 79]. A dedicated CR (inverted elliptical (m_{HH}) mass cut, Eq. 22) with $< 7\%$ contamination; COMBINE fit with Barlow–Beeston gives ttSF $\approx$ 0.91–0.99 [CMS p. 81,82]; the shape is taken from MC (no shape dependence observed).
- **Jet$\to$ τ fakes — data-driven** [CMS p. 94,95]. $> 93\%$ of the events with a jet faking a τ; fake-rate uncertainty per region (barrel 24/13/11%, endcap 28/18/25% by year, Table 33), applied as an event weight.

- **Single Higgs — irreducible, same final state** [CMS p. 82]. ggH, VBFH, VH, $t\bar{t}H$ with $H \to b\bar{b}$ or $\tau\tau$ mimic the signal; modelled from MC. (Note: $t\bar{t}H/VH$ couplings make this a *coupling-dependent* background.)
- **Di-/tri-boson, W+jets, single-t, V+2jets — MC** [CMS p. 82].

Why this is a new, NSBI-specific difficulty

1. **No density for data-driven backgrounds.** QCD (ABCD) and jet$\to\tau$ fakes are estimated as *yields/weights from data control regions*, not as simulated events with a per-event probability. NSBI's per-event likelihood ratio $P_X(x)/P_{\text{ref}}(x)$ is undefined for them. Folding an ABCD/fake estimate into an *unbinned* per-event likelihood is an open problem the off-shell analysis never had to solve (its one big background was MC). Options (all imperfect): treat QCD/fakes only in a binned auxiliary term; build a surrogate density from the control-region data; or restrict NSBI to the MC-modelled part and bin the rest.
2. **The QCD shape systematic is a top-ranked nuisance** [CMS p. 91] — and it is intrinsically *asymmetric/ill-defined* (one alternative template used for both up and down), forcing a symmetrisation hack [CMS p. 96,97]. Carrying that into a per-event morphing is non-trivial.
3. **Single Higgs is an irreducible, coupling-dependent background** in the same final state — it must be modelled *and* morphed consistently with the signal couplings, with its own (correlated) theory systematics.
4. **Many background-normalisation NPs are themselves data-fit results** (18 DY SFs with a covariance, ttSF, QCD B/D) — these are constrained *in situ* in the binned CMS fit; reproducing that constraint inside an unbinned NSBI fit requires the CRs to be represented in the per-event likelihood too.

A workaround: flow-based data-driven backgrounds (ABCDnn)

The "no per-event density" problem is the sharpest objection to NSBI here, but it is *not* a dead end — there is a demonstrated solution. **ABCDnn** (Choi, Lim, Oh, arXiv:2008.03636) generalises the ABCD method with a *neural autoregressive flow* that learns the full multi-dimensional CR$\to$SR transformation, conditioned on the discriminating observables. Where standard ABCD gives only a scalar SR yield ($N_D = N_B N_C / N_A$), ABCDnn gives a full *differential* background distribution $p_{\text{QCD}}(x)$ — which can then be turned into a density ratio $r_{\text{QCD}}(x) = p_{\text{QCD}}(x)/p_{\text{ref}}(x)$ by training a classifier between flow-generated events and the pooled reference, so it enters the NSBI likelihood *on the same footing* as the MC-derived ratios.
The concrete production example (S. An, Dutta/Paulini, CMU PhD thesis 2025, Ch. 5): the CMS *all-hadronic four-top* search, Run-2 ($137\,\text{fb}^{-1}$, 13 TeV) + Run-3 ($62\,\text{fb}^{-1}$, 13.6 TeV), reaching an **observed (expected) significance of** $3.95\,(1.65)\,\sigma$, $\mu = 2.7^{+1.1}_{-0.9}$. Its single most important design choice is a **yield/shape split** of the dominant $t\bar{t}$+jets and QCD backgrounds:

- *Yield* ← **ExtendedABCD**: standard ABCD in (N_{jets}, N_b) *fails* here (the factorisation breaks at high multiplicity — off by ∼18–19.5% in closure); adding two extra sidebands restores it to ∼1–7% (their $\hat{F}_D = (F_B F_C/F_A)^2(F_X/F_B F_Y)$, Table 5.1–5.2). The lesson: *get the normalisation from a robust closed-form estimator with enough sidebands, not from the flow.*
- *Shape* ← **ABCDnn** (a neural autoregressive flow): it learns a transformation that *morphs the $t\bar{t}$ MC into the data-driven $t\bar{t}$+QCD estimate* (target = data − signal − minor bkg per CR), trained by minimising **Maximum Mean Discrepancy** and *conditioned on the CR*

position (N_{jets}, N_b), then applied to $t\bar{t}$ MC in the SR. Notably it never uses QCD MC (too few SR events) — the flow turns pure-$t\bar{t}$ MC into an effective $t\bar{t}$+QCD mixture.

Three training tricks they found necessary (worth copying): **loss-weighted CR sampling** (sample harder CRs more often, $\propto e^{\text{loss}}$, so every region is learned uniformly); **continuous** (N_{jets}, N_b) **conditioning** instead of one-hot labels, so the flow *interpolates smoothly* across CRs — this is what makes the CR→SR *extrapolation* well-behaved; and a **warm-up** (pre-train near identity, RealNVP-style) for stable convergence.

Application to $b\bar{b}\tau\tau$: the **jet→ τ fakes** are the natural target — a fake-factor method is conceptually an ABCD with τ-ID inversion as the second axis, so an ABCDnn conditioned on $(p_{\text{T}}^{\tau}, \eta^{\tau})$ could learn the full fake-τ kinematic shape instead of the current coarse 2D fake-factor binning; the QCD multijet is the other target; and for $t\bar{t}$ the flow can serve as a data-driven shape cross-check. Following the four-top lesson, keep the **yield/shape split**: take the fake-τ/QCD *normalisation* from the existing fake-factor/ABCD machinery (which CMS already trusts, [CMS p. 61,94]) and use ABCDnn only for the differential *shape*, then convert that shape to a per-event ratio $r(x)$ for the likelihood. A fully realised $b\bar{b}\tau\tau$-NSBI would use *ABCDnn shapes (with ABCD/fake-factor yields) for QCD/fakes, MC ratios for the MC-modelled components (Z+jets, $t\bar{t}$, single-H, diboson), and one unbinned NSBI likelihood combining both* — removing the main caveat that NSBI "depends on MC where MC is wrong."

But the generic caveats remain: (i) *extrapolation distance* — the CR→SR map strains for tight SRs far from the sidebands in tagger/isolation space (multiple overlapping CRs validate it but cost stats); (ii) *uncertainty propagation* — the four-top analysis does *not* bootstrap the flow; it splits the uncertainty into three data-driven nuisances (thesis Sec. 6.1, Table 6.1): the *finite-CR-statistics* piece is a lnN normalisation NP propagated through the ABCD formula ("statistics of extended-ABCD," 4–26%); the *normalisation-modelling* piece is a lnN NP from a validation-region (VR) data-vs-prediction closure (weighted-mean deviation $1 - \langle f \rangle$ + weighted RMS of $f = N_{\text{data}}/N_{\text{pred}}$, 6–44%); and the *ABCDnn shape* uncertainty is a shape NP implemented as a *linear shift of the discriminant*, with the shift size chosen by *minimising a quantitative data/pred-agreement metric* in the VR (floor 1%, correlated across H_T, uncorrelated across years). The whole uncertainty-*estimation* procedure is itself validated in a dedicated orthogonal "closure-test region" that checks pred+syst covers truth in the SR. *Caveat for NSBI:* this VR-closure shape nuisance ("shift the discriminant") is a *binned-fit* construct; in an unbinned NSBI it becomes yet another per-event morphing nuisance on the ratio $r(x)$ — i.e. the same learned-morphing problem, now applied to the data-driven component; (iii) *signal contamination in CRs* — if signal leaks into the control region the flow learns "background + signal" and biases the SR low, requiring iterative CR definitions with signal subtraction (this caveat is *mild* for $b\bar{b}\tau\tau$: the HH signal is tiny in the fake-τ/QCD control regions, just as the SM four-top signal was negligible in its CRs); (iv) *no theory handle* — the flow gives a shape but no ME+PS+PDF decomposition, so generator-level theory uncertainties on the background cannot be propagated and are replaced by a data-driven CR→SR closure systematic.

The one caveat specific to using ABCDnn *for NSBI* (the real gap). In the four-top thesis the flow models only a *low-dimensional* target — the event-level BDT discriminant plus H_T (two variables) — because the downstream fit is a *binned template* fit of that discriminant. NSBI needs the opposite: the *full per-event density* over the whole feature vector, so the data-driven shape can become a per-event ratio $r(x)$. So the four-top result proves flows can deliver a data-driven background *shape of a summary observable*; extending that to the full-dimensional per-event density an unbinned NSBI consumes is a genuine step beyond what is demonstrated — a higher-dimensional flow, harder to train and validate. The conservative intermediate is to let ABCDnn

produce the data-driven shape of the *NSBI discriminant* (the per-event log-ratio summed over components), i.e. treat QCD/fakes as a binned component of an otherwise-unbinned likelihood — exactly the "keep QCD binned, NSBI-fy the MC components" compromise.
Net: ABCDnn downgrades the data-driven-background problem from "open" to "solved-with-caveats" for a binned *discriminant shape; the fully-unbinned version is a worthwhile but unproven sub-project for a $b\bar{b}\tau\tau$-NSBI pilot.*

5 Systematics handling

What is genuinely shared (equal for both, and equally fallible): the *sources and* $\pm 1\sigma$ *sizes* of the NPs; the *object-level propagation* (shift the object and all dependents — e.g. $\tau \Rightarrow$ MET$\Rightarrow$SVfit — and recompute the discriminant, [CMS p. 93]); the *factorisation over NPs* and $\pm 1\sigma$ *interpolation*; *Gaussian-constrained profiling*; and the κ_λ*-dependent theory* treatment. ATLAS builds the per-event systematic weight as $P_X(x,\alpha) = w_X(x,\alpha)\, P_X(x,0)$ with $w_X(x,\alpha) = \prod_m w_X(x,\alpha_m)$, each single-NP weight interpolated poly-exp between $\pm 1\sigma$ classifier estimates [ATLAS p. 83,84]. CMS does the per-*bin* analogue (re-run the DNN on varied inputs $\rightarrow$ template).

What is harder for $\kappa_\lambda/b\bar{b}\tau\tau$:

1. **κ_λ–systematic shape degeneracy.** The dominant experimental systematics (JES, τES, MET, signal parton shower) deform *the same m_{HH} shape* that carries κ_λ. In 4ℓ the big systematics (theory) are largely *separable* from the signal shape; in $b\bar{b}\tau\tau$ they are not. Decorrelation is therefore impossible (it would erase the signal), forcing the "model-it" route — so the accuracy of the morphing $\Delta(x|\alpha)$ *directly sets* the κ_λ uncertainty. (A Fisher-information toy gives, schematically, a $\sim 13\times$ systematic inflation from m_{HH} alone collapsing to $\sim 1.3\times$ with the full feature set — the degeneracy is broken by dimensionality, not avoided.)

2. **An order of magnitude more, and larger, experimental NPs** [CMS p. 90–97]: JES split into a reduced set of **11 sources** with a year-correlation scheme [CMS p. 95]; τ**ES** p_T-dependent with 4 decay-mode components [CMS p. 94]; ***b*-tag** 5 components (hf, lf, hfstats, lfstats, cferr) [CMS p. 96]; **DeepTau** vsJet (5 p_T bins) + vsEle (2 η bins) [CMS p. 95,96]; trigger SFs, JER, PU-jet-ID, L1 prefiring, $e/\mu \rightarrow \tau$ fake ES. The off-shell analysis has comparatively few, small (lepton) experimental NPs. Each NP is another per-event weight to learn ($\sim$2000 NNs already in off-shell = 2×NP×process; $b\bar{b}\tau\tau$ grows this).

3. **Data-driven background systematics that don't fit the density-ratio picture**: the QCD norm (5–100%) and high-ranking QCD shape [CMS p. 91,92,96,97], the 18 DY SF shape/norm templates [CMS p. 94], the jet$\rightarrow \tau$ fake-rate uncertainties (11–28%, Table 33) [CMS p. 95], the VBF dipole-recoil PS uncertainty (20–70%, Table 32) [CMS p. 93]. These are template/weight constructs, not deformations of an MC density.

4. **κ_λ-dependent theory uncertainty (POI×systematic entanglement).** The HH scale/PDF/m_t uncertainties are folded into normalisation NPs that are strongly κ_λ*-dependent*: Run-2 quoted `THU_HH` $= +4\%/-18\%$ at the SM [CMS p. 91]; the Run-3 note (AN-25-103) gives ggF scale+m_t-scheme $= +6/-23\%$ at the SM and tabulates the band *per κ_λ point*, with the m_t asymmetry *flipping sign* across the scan (its Table 40). The off-shell μ does not entangle with its theory uncertainty this way. A faithful *per-event* treatment would require POI-coupled morphing — but see the AN-25-103 reassessment below: the *rate* part is handled pointwise without any such object; only the *shape* part remains open.

5. **Factorisation is more likely to break.** $w_X = \prod_m w_X(\alpha_m)$ assumes NPs are independent sources [ATLAS p. 83]; but JES and τES *both* feed m_{HH} $(= m(b\bar{b}+\tau\tau))$, so a genuine cross-term exists in $b\bar{b}\tau\tau$ that is benign in clean 4ℓ.

6. **MC-statistical uncertainty of the learned ratios.** ATLAS propagates it as one NP via the NN-ensemble spread [ATLAS p. 86,87]; CMS uses Barlow–Beeston per bin. With far smaller *systematic-variation* samples in $b\bar{b}\tau\tau$, this term is more important (the nominal MC, ≈500k–750k/κ_λ-point, is ample for training; the binding constraint is the least-populated $\pm 1\sigma$ sample). *Candidate estimators* (the choice to settle before quoting a number): the ATLAS NN-ensemble-spread/double-bootstrap (argued *incorrect* by Shyamsundar, 2505.19156); and the analytic-frequentist **wifi ensembles** (Benevedes & Thaler, 2506.00113) — a basis-function expansion whose weight covariance propagates in closed form to a per-event uncertainty band on $r(x)$, with demonstrated coverage on an SBI density-ratio example and *no bootstrap*, so it is attractive on both correctness *and* the ensemble/compute cost (untested on interference/morphing/weighted-MC). The wifi paper's own coverage tests add a *second, empirical* strike against the spread-style estimator: uniform-weight ("naive") ensembles come out *biased and under-covering*. Full dissection of the ATLAS estimator and the wifi plug-in map: the item-6 update below.

Two systematics that NSBI actually handles *better* than templates. Worth stating the upside too: (a) *parton-shower / generator* uncertainties (Pythia vs. Herwig, ISR/FSR) are a natural fit — a **CARL/DCTR**-style reweighting network learns the per-event weight between alternative showers, which is then profiled as a nuisance; this is strictly more powerful than the current "shape envelope over generator variations" and is the in-likelihood cousin of the RS3L robust-representation idea. (b) b-tag / τ-ID scale factors can be *absorbed* into the density ratio and calibrated in situ against a standard candle (the $VZ(c\bar{c})/Z \to b\bar{b}$ peak trick), turning the POG SF program into an external cross-check rather than the primary constraint. So the systematics ledger is not all cost: *the morphing-heavy ones (JES/τES on m_{HH}) are the burden; the modelling ones (shower, flavour-tagging) are where NSBI's per-event treatment is an asset.*

Literature maturity & risk (where each challenge actually stands). A targeted sweep of the 2024–2026 literature shows that *none* of the six difficulties above is cleanly "numerical" (method exists, just measure how bad): the $b\bar{b}\tau\tau$ regime consistently exceeds where any per-event method has been demonstrated. The largest published demonstrations of *learned* NP morphing are ∼3 calibration NPs on *simulated* data (GOLLUM; Schöfbeck *et al.*, PRD 112 052006) and ∼50 NPs on real but *clean* 4ℓ data (ATLAS, 2412.01600). Nothing exercises the actual $b\bar{b}\tau\tau$ regime: ∼40 *shape-degenerate* experimental NPs deforming a *reconstructed* m_{HH} across three channels with data-driven backgrounds.

Challenge (item above)	Literature maturity	Class	Risk
1 κ_λ–syst. shape degeneracy	poly-exp morphing exists (ATLAS, GOLLUM); per-event closure under *simultaneous* JES$\times\tau$ES on m_{HH} unreported	hybrid	high
2 order-of-mag. more NPs ($\sim$40)	$\leq$50 NPs on clean 4ℓ (ATLAS); $\leq$3 calib. NPs on sim data (GOLLUM); cross-terms + calibration at this scale unproven	hybrid	high
3 data-driven bkg systematics	components exist (ABCDnn, neural FF, GOLLUM) but *never combined* per-event; neural FF (2511.06972) explicitly defers systematics	hybrid (borderline open)	high
4 κ_λ-dependent (POI-coupled) theory	*split* (AN-25-103): *rate* band = standard practice (pointwise κ_λ-dependent lnN, its Table 40); per-event *shape* part = proposed only, done by nobody	**C4a closed / C4b open (upgrade)**	med
5 factorisation cross-term (NP$\times$NP)	per-event *toy* only (GOLLUM, sim data); τES$\to$MET$\to$SVfit$\to m_{HH}$ recompute untested	hybrid	med
6 MC-stat of learned ratios	real (ATLAS) but *two strikes*: correctness critique (Shyamsundar, 2505.19156; $\sim$30$\times$ toy underestimate) + empirical naive-ensemble undercoverage (2506.00113 tests); leading fix = *wifi ensembles*	hybrid	high
a PS/generator via CARL/DCTR	components exist; never demonstrated end-to-end as a profiled NSBI nuisance	hybrid	med
b absorb b-tag/τ-ID SFs	building blocks only; no per-event SF-absorption + in-situ standard candle shown	hybrid	med

Open vs. numerical — the verdict

One challenge was rated conceptually open — and it has since split (see the AN-25-103 reassessment below): #4, the κ_λ-dependent / POI-coupled theory morphing $w_\theta(x;\kappa_\lambda,\alpha_\theta)$. The standard factorised scheme keeps every NP weight *POI-independent* ($w_m(x;\alpha_m)$), and off-shell avoids the problem entirely (μ does not couple to its theory band because its signal *shape* is a fixed mixture). Note #4 (POI$\times$NP) is distinct from #5 (NP$\times$NP): GOLLUM solves the latter on a toy but does *not* reach the former. *Post-AN-25-103*: the *rate* half (C4a) is **closed** — a pointwise κ_λ-dependent lnN, standard CMS practice — while the per-event *shape* half (C4b) remains open but is an *upgrade* that no analysis performs. #3 (per-event systematics of a *data-driven* density) is the borderline second — every component exists but none combine. The other six are *hybrid*: a method exists in an easier setting, so each "pilot" is a *method-selection gate*, not pure quantification. **Most dangerous to the headline** (a *tighter* interval that must still *cover*): #6 — Shyamsundar (2505.19156) argues via a toy that the ATLAS double-bootstrap MC-stat estimator does *not* capture the finite-MC uncertainty — and uniform-weight ensembles under-cover *empirically* in 2506.00113's own tests — so settle this before quoting a number (leading candidate replacement: the analytic-frequentist *wifi*-ensemble estimator, 2506.00113); and #2 — the $\sim$40-NP scale may force a conditional surrogate (GOLLUM/R4) or GP morphing (R3) over $\sim$thousands of classifiers. **Suggested gates first:** #4 (does a POI$\times$NP ansatz work, on a toy) and #6 (which MC-stat estimator is correct); then the #2 scaling gate; then #3/#1/#5 closures; the upsides (a,b) are field-first demonstrations. *(Gate #4 refocused post-AN-25-103: the rate half is closed; the pilot is now the theory-*shape *truncation-bias measurement — see the reassessment below.)*

Why off-shell escapes #4, and the likely way out. Algebraically off-shell and κ_λ are *identical* — POI coefficients on fixed templates, $(\mu-\sqrt{\mu},\ \sqrt{\mu},\ 1-\sqrt{\mu})$ vs. $(\kappa_\lambda^2,\ \kappa_\lambda,\ 1)$ — so the difference is not the POI structure but *what the theory uncertainty acts on.* THU$_{HH}$ sits on the *signal*, and the triangle/box/interference carry *different* scale/PDF/m_t bands; κ_λ sets their mixture, so the net theory deformation's size *and* its m_{HH} shape vary with κ_λ (the *fractional* band even diverges near the $\kappa_\lambda \approx 2.45$ cross-section minimum, where triangle and box cancel). In off-shell the leading theory NP is the $gg \to ZZ$ continuum K-factor — on the *background* (coefficient 1, μ-independent) — and σ is monotonic in μ with no cancellation, so μ never re-mixes a component-differential theory band. **Way out (first thing to try):** replace the single total-signal THU$_{HH}$ NP with *per-component, POI-independent* theory weights $w_\triangle$, w_{int}, $w_\square$ (functions of α_θ only); the κ_λ coefficients then blend them and the κ_λ-dependence of the net THU emerges *automatically*, preserving factorisation. The two missing pieces (hence "open"): the *component-differential* theory bands (the WG tabulates only the total band per κ_λ point), and any NSBI closure test of per-component theory morphing. *This prediction is now empirically confirmed — see the sign flip in the AN-25-103 update just below.*

Update (12 Jul 2026): how the theory uncertainty is *actually* handled in the current CMS $b\bar{b}\tau\tau$ analysis (Run-3, AN-25-103) — and the C4 reassessment. Reading the Run-3 note confirms the theory uncertainty is the *leading* $b\bar{b}\tau\tau$ systematic and reveals a three-layer treatment:

1. *POI rate+shape, by construction.* The ggF signal at any $(\kappa_\lambda, \kappa_t)$ is an *exact* three-component template combination, $\sigma(\kappa_\lambda, \kappa_t) = \mathbf{k}^{\mathrm{T}}\mathbf{K}^{-1}\boldsymbol{\Sigma}$, built from three NLO POWHEG samples at $\kappa_\lambda =$ 1, 2.45, 5 and applied *per bin at fit time* (AN Eq. 52–55; the VBF analog uses six components). This is precisely the mixture / known-green-coefficient structure of Sec. 1, executed on templates.

2. *Theory NPs are pure normalisation lnN.* QCDscale, PDF and α_s enter as rate modifiers with log-normal priors (AN Sec. 7.1, Table 42): at the SM point, ggF scale+m_t-scheme = $+6/-23\%$ (the -23% dominated by the m_t*-scheme* choice, not scale), PDF $\pm1.5\%$, α_s $\pm1.7\%$; VBF scale $+0.05/-0.03\%$, PDF+α_s $\pm2.7\%$. **No theory-shape NP exists for the signal** — scale/m_t variations never deform the templates.

3. *The κ_λ-dependence is handled at interpretation time.* AN Table 40 tabulates the band *per κ_λ basis point* — large and *sign-flipping*:

κ_λ	$\sigma(\kappa_\lambda)$ [fb]	scale	m_t scheme
1	34.13	$+2.1/-4.9\%$	$+4/-18\%$
2.45	14.92	$+3.2/-5.5\%$	$+4/-22\%$
5	99.65	$+7.9/-8.4\%$	$+\mathbf{13}/-\mathbf{4}\%$

The scans are *pointwise*: at each fixed κ_λ hypothesis the model is *rebuilt* (templates via Eq. 55, band at that κ_λ) and all nuisances profiled (AN Sec. 8.4.5); 2-D exclusions compare the rate limit to σ_{theory} *evaluated at each parameter point*, with the band propagated to the exclusion boundaries (Sec. 8.4.6); the Run-2+3 combination treats the joint QCD+m_t ggF uncertainty as *correlated* across runs (Sec. 8.5.1).

Concept: what a "scan" actually is — and why κ_λ can be fixed inside every fit

A profile-likelihood "scan" is not one fit: it is a **grid of conditional fits, one per hypothesis value**. Each fit answers one frozen question — *"if κ_λ were exactly this value, how well can the model explain the data, with every nuisance doing its best?"* — and returns *one number*, the profiled NLL: one evaluation of the numerator of $\lambda(\kappa_\lambda) = L(\kappa_\lambda, \hat{\hat{\alpha}}(\kappa_\lambda))/L(\hat{\kappa_\lambda}, \hat{\alpha})$. The "scan" is the assembled table of (κ_λ, t) pairs; $\hat{\kappa_\lambda}$ is the *curve's* minimum and the CI its $t \leq 1$ crossings — read off the assembly, never from a single fit. (Valley picture: each fit is a probe dropped to the valley floor at one longitude; no probe moves sideways, yet the probe depths trace the floor-vs-longitude curve.) **Why pointwise** rather than one fit with κ_λ floating: (i) a free fit yields only the minimum and a Hessian *parabola* — blind to the grossly non-parabolic κ_λ curve (the AN's own scan, its Fig. 131, shows the interference structure near $\kappa_\lambda \approx 5$–6); (ii) each minimisation is over nuisances only, hence robust; (iii) **the model may be rebuilt per point** — re-mixed templates *and* that point's theory band — because the minimiser never needs $\partial(\text{model})/\partial\kappa_\lambda$: the κ_λ-dependence of the band never enters any derivative, pull, or profiling step. It is bookkeeping *between* fits, not structure *within* one. **NSBI scans are pointwise in exactly the same way** (the prelim's profile-NLL scan already evaluates $t(\kappa_\lambda)$ on a grid over cached ratios), so the identical trick ports verbatim. Mild costs, engineering not conceptual: grid resolution (interpolate near thresholds and the σ-minimum) and per-point convergence.

C4 reassessed (12 Jul 2026): it splits — the rate part is closed, the shape part is an upgrade

C4a — the κ_λ-dependent *rate* band: closed (numerical, portable today). Because any scan (binned *or* NSBI) profiles at fixed κ_λ pointwise, a κ_λ-dependent lnN *magnitude* needs no POI-coupled per-event object: put band(κ_λ) (LHC HH WG / AN Table 40) into the green rate factor on $\nu_{\mathrm{sig}}(\kappa_\lambda)$, profiled per scan point and correlated across points/runs exactly as CMS does. Since the theory uncertainty is the *leading* systematic and the rate band carries the -23%, **the dominant $b\bar{b}\tau\tau$ systematic is implementable in the NSBI fit at parity with CMS, today. C4b — the theory *shape* deformation: still open, but an *upgrade*, not a parity risk.** Scale/m_t variations deform each component's $d\sigma/dx$ (hence the discriminant) — the per-component $w_{\theta,s}(x;\alpha_\theta)$ morphing that *nobody* performs: CMS is normalisation-only. So C4b is not a gap NSBI must close to match the baseline; it is where *both* approaches currently truncate, and where NSBI's per-component structure is the natural upgrade. Table 40's **sign flip** ($+4/-18\%$ at κ_λ=1 $\rightarrow$ $+13/-4\%$ at κ_λ=5) is direct empirical evidence for the per-component picture: the net band rotates because the components' bands differ — exactly what blending POI-independent component bands with the known κ_λ coefficients reproduces automatically. **Gate refocus + recommended default:** adopt **CMS-parity as the default for the headline comparison** — C4a only (rate-band lnN per scan point, correlated across runs), *no* theory-shape nuisance — so the NSBI-vs-binned CI comparison holds every systematic fixed and attributes the difference to *unbinning alone* (this also preserves the best-case width guarantee of the width analysis below, and is the review-defensible choice: "our systematic model is identical to AN-25-103, NP for NP"). The truncation risk is then *shared* with the baseline — but possibly not *equally* shared: an unbinned fit reads shape at full per-event resolution and may convert an unmodelled theory-shape distortion into a *larger* κ_λ bias than the binned fit (or a smaller one — in the full feature space the distortion may look *less* like κ_λ; empirical either way). Hence the pilot is **two-sided**: generate Asimov data with scale-varied *shapes* (free — μ_R/μ_F variations are stored per-event generator weights, no new samples), fit *both* the binned and the NSBI models in the shared normalisation-only configuration, and compare κ_λ biases. Both small $\Rightarrow$ parity validated; comparable $\Rightarrow$ shared caveat, comparison still clean; NSBI visibly larger $\Rightarrow$ C4b graduates from optional to needed-for-our-method. Either way, **C4b is a separable Phase-3 *methods* upgrade** — a *fidelity/coverage* improvement, not a CI-tightening one (it likely costs width, though a shape NP is partially data-constrainable in situ, unlike a signal-rate lnN) — natural in the per-component architecture, cheap (scale weights need no new samples), and never a prerequisite for the headline result (folded into studies #5/#19).

Update (12 Jul 2026): training statistics of the learned ratios (item 6) — the ATLAS double-bootstrap dissected, and the wifi map. Two distinct "statistical" uncertainties must not be conflated: the *data* statistics (Poisson fluctuations of the observed events — dominates the CI in this stat-limited analysis, is handled exactly by the likelihood, and is where NSBI's per-event *gain* lives) and the *training* statistics (finite MC $\Rightarrow$ an error on every *learned* ratio — item 6). The second is existential for the headline claim: a *tighter interval from the same data* is only credible if the learned-ratio noise is quantified with correct coverage — and it concentrates exactly where the samples are smallest (the $\pm1\sigma$ variations).

What ATLAS actually does (off-shell [ATLAS p. 86,87], as dissected in 2505.19156). Two bootstrap levels: (1) resample the training set D M times, train one NN per replica, and *uniformly average* the M log-ratios into the estimator; (2) resample *the ensemble itself* K times with replacement and take the spread of the K re-averaged estimators as the uncertainty, inserted into the fit as a single coherent (spurious-signal-style) nuisance. *The critique:* level 2 resamples *conditional on the fixed* D, so it estimates only the finite-*ensemble* noise, $\mathrm{Var}[\mathrm{avg} \mid D] \sim 1/M$ — which vanishes as $M \to \infty$ at fixed N — and *not* the finite-*sample* deviation of the trained estimator from the true ratio ($\propto 1/N$, independent of M). In Shyamsundar's Gaussian toy the true uncertainty is $\sim$0.1 while the double-bootstrap reports $\sim$0.003 — a $\sim$30× underestimate. *Second, independent strike:* the wifi paper's own coverage tests find uniform-weight ("naive") ensembles — structurally the ATLAS aggregation — *biased and under-covering*, while their estimator covers. (Fairness note: in the off-shell regime the MC was ample relative to the data sensitivity, so the practical impact there may have been limited; in $b\bar{b}\tau\tau$ the $\pm 1\sigma$ samples are the *binding* constraint, so the estimator cannot be inherited on trust.)

What wifi does instead (2506.00113; mechanics verified). The name is the construction: $w_i f_i$. (1) Train M networks and *freeze* them as basis functions $f_i(x)$ — for us, the already-trained prelim ensembles; basis diversity via their *Bootstrap protocol* (members trained on bootstrap replicas — i.e. ATLAS's level 1, reused). (2) Re-fit the *linear* combination $\log \tilde{r}(x|w) = \sum_i w_i f_i(x)$ on a *held-out* split with a symmetrised, normalisation-enforcing classifier loss — a *convex* problem. (3) $\mathrm{Cov}(\hat{w})$ from the **sandwich estimator** $C = V^{-1} U V^{-1}$ (M-estimator theory: Hessian V, score covariance U — anchored to the *sample*, not the ensemble). (4) Propagate *analytically*: per-event band $\sigma^2(x) = f_i(x)\, C_{ij}\, f_j(x)$, and through a downstream fit via the **Gong–Samaniego** closed form, $\sigma^2_{\mathrm{GS}} = \sigma^2_{\mathrm{MLE}}\,(1 + \sigma^2_{\mathrm{MLE}}\, A_i C_{ij} A_j)$ — no extra NP, no toys, no bootstrap retrainings. Demonstrated nominal coverage on a Gaussian toy ($M \geq 8$) and on q/g-jet SBI ($M = 32$, Bootstrap protocol). **The structural relation in one line:** both start from the same level-1 bootstrap ensembles and diverge *after* — uniform average + resample-the-ensemble spread (wrong target, $\to 0$ as M grows) vs. held-out convex re-fit + sandwich covariance (right target, anchored to N).

	ATLAS off-shell (double-bootstrap)	**wifi ensembles**
target of the estimate	$\mathrm{Var}[\text{ensemble avg} \mid D]$ (level-2 spread)	asymptotic variance of the fitted ratio over data realisations (sandwich)
scaling	$\sim 1/M$ at fixed N ($\to 0$: the pathology)	anchored to N via the score; M only enriches the basis
aggregation	uniform average of members	optimal convex re-fit of w on a held-out split
enters the fit as	one coherent NP (spurious-signal style), profiled	closed-form Gong–Samaniego inflation per scan point (no NP)
granularity	single global shift	full functional covariance via the basis
cost	M trainings (+ cheap resampling); the *corrected* bootstrap needs $M \times K$ retrainings	M trainings (reused) + one convex fit; no retraining multiplier
validation status	internal calibration; toy counterexample ($\sim$30× under); naive-ensemble undercoverage (2506.00113 tests)	nominal coverage on toy + q/g; *untested* on interference / weighted MC
key assumptions	level-2 bootstrap validity (contested)	well-specified basis; asymptotics; *unweighted* events

The concrete plug-in map (five points).

1. *Family-1 ratios:* wrap the *existing* prelim ensembles as the frozen basis; convex re-fit on a held-out split; sandwich $C \Rightarrow$ a calibrated per-event stat band on every $r_s(x)$ at near-zero extra training cost.

2. *Honest morphing (#9):* the same band on each Family-2 anchor classifier is the *statistical* part of the morphing uncertainty — exactly where the $\pm 1\sigma$ samples are smallest.
3. *R3 synergy:* the anchor bands enter the GP interpolation as *heteroscedastic noise* — wifi handles within-anchor training stat; the GP handles between-anchor interpolation.
4. *The fit:* propagate C through the pointwise JAX scan via Gong–Samaniego (A from autodiff) $\Rightarrow$ the paper-ready decomposition "training statistics contribute $X\%$ of $\sigma(\kappa_\lambda)$."
5. *The C6 gate, precisely labelled:* a three-way coverage shootout on $b\bar{b}\tau\tau$ toys — ATLAS-style naive spread vs. the *corrected* bootstrap (Shyamsundar's fix: fresh ensembles per pseudo-experiment, $M \times K$ retrainings) vs. wifi — at nominal and $\pm 1\sigma$-mimicking downsampled statistics.

Caveats before leaning on it: **weighted/negative-weight events are not addressed in the wifi paper** (our NLO samples have them; the weighted-M-estimation extension is ours to derive and validate — the genuinely new piece); the *well-specified-basis* assumption means the covariance sees variance around the basis-restricted optimum, not a shared blind spot of all members — C2ST closure stays mandatory; asymptotics must be checked at the smallest $\pm 1\sigma$ samples; budget $M \approx 16$–32 (Bootstrap protocol — their Partition protocol degraded in high dimension); and the interplay with the isotonic-calibration step is an implementation detail to settle. Study #15 is rewritten as this pilot. A Phase-2 crossover: frozen foundation-model embeddings as a *shared fixed basis* + linear heads make every ratio a GLM with exact sandwich covariance — frozen-FM was worst for point-estimate *separation*, but may be ideal as a *covariance basis*.

Where the impacts actually land (AN-25-103 Fig. 118: pulls/impacts on the Asimov dataset, POI = r multiplying SM ggHH+qqHH, full Run-3). The total Asimov uncertainty is $r = 1.00^{+1.69}_{-1.37}$ — strongly stat-dominated, consistent with the expected limit of 3.3×SM. The leading nuisances (bar lengths read off a draft plot; treat as ±0.05-level approximate):

Rank	NP	Impact $\lvert\Delta r\rvert$	Prior (input) size	Challenge row (Sec. 5 matrix)
1	`CMS_scale_j_2024` (JES, 2024 era)	$\approx +0.5/-0.2$	$\pm 1\sigma$ jet calibration (2024 JEC still preliminary)	**C1** (deforms m_{HH}/DNN inputs) + one of C2's $\sim$40
2	`QCDscale_ggHH+mtop` (theory)	$\approx \mp 0.3$–0.35	$+6/-23\%$ at SM (Table 40; sign-flips with κ_λ)	**C4** (POI-coupled theory; C4a rate half)
3–4	`MC stat`, $\tau\tau$ res2b 2024, bins 3/3, 2/3	$\approx \pm 0.25$–0.3 *each*	per-bin template Poisson (large in sparse high-score bins)	**C6** (binned analog of learned-ratio training stat)
5	`bbtt_norm_ggH` (single-H normalisation)	$\approx \pm 0.27$	not stated in the pages read	none of the hard rows — benign green-factor lnN (study #12)
6–8	more MC-stat bins; `CMS_res_j_2024` (JER)	$\approx \pm 0.2$–0.25	—	C6; C1/C2

Three structural observations. (i) **Roughly half of the top-25 rows are individual MC-stat bins**; their naive quadrature aggregate ($\sim$0.6–0.8, with more bins in ranks 26–75) makes finite-simulation statistics arguably the *largest single systematic block* in the current analysis — which the AN itself concedes (Sec. 7.3: MC statistics is "the limiting factor" for shape systematics, forcing symmetrisation/envelope manipulations). The binned baseline is *already* MC-stat-dominated; the NSBI version of that block is exactly the learned-ratio training stat (item 6/wifi). (ii) The impact spectrum *drops an order of magnitude* after the top-25 (Fig. 119 spans only $\lvert\Delta r\rvert \lesssim 0.06$): jet$\to\tau$ fakes (`CMS_FF_syst`, rank $\sim$33) and luminosity (rank $\sim$40) are mid-list, *not* leaders. (iii) The empirical top — {JES, theory, MC-stat} — is precisely {C1, C4, C6}, the three challenges this note's risk matrix prioritises, now confirmed by the analysis's own impact ordering.

Why "the leading systematic is theory" is consistent with JES topping this plot. Four reconciliations, all real: (i) *grouping granularity* — JES is split into per-era NPs (2022, 2022EE, 2023, 2023BPix, 2024, plus JER), each ranked separately, while the QCD+m_t theory band is *one* NP; "leading systematic" statements usually refer to *grouped* uncertainty breakdowns, not single-NP rows. (ii) *Correlation structure* — experimental NPs are uncorrelated across eras and across Run-2/Run-3 (different detector conditions, Sec. 8.5.1), so in combinations they average down alongside statistics; the theory band is *fully correlated* across everything (Sec. 8.5.1) and is therefore a *floor* whose relative share grows with luminosity. (iii) *POI context* — for the κ_λ interval and the $\sigma\times$BR-vs-theory crossing, the theory band enters *directly* through $\sigma_{\text{theory}}(\kappa_\lambda)$, largest ($-22\%$) exactly near the σ-minimum region that controls the interval edges; JES never touches σ_{theory}. (iv) *Irreducibility* — a signal-normalisation theory NP is degenerate with r (no shape handle; the data cannot constrain it), whereas JES is constrained in situ and its post-fit impact shrinks as data grow. A practical footnote: the 2024-era JES sits at rank 1 partly because the 2024 jet calibration is still preliminary in this draft — expect it to deflate by unblinding. *Caveats: draft, blinded (Asimov), POI = r not κ_λ; in a κ_λ scan the C1 (JES/τES) and C4 (theory, via the sign-flipping band) entries would if anything climb; the rank-5 label is small in the plot — re-verify at unblinding.*

Best-case CI *width* vs. the binned fit (a complementary axis). The matrix above asks *can we do it*; this asks *what it buys on the interval width if we do* — assuming the morphing is calibrated and coverage correct (i.e. setting aside item 6's correctness question). Two baselines must be kept apart. *Relative to the stat-only NSBI number* (the preliminary $2.75\sigma \to 7.1\sigma$), honestly including systematics widens the interval for *both* NSBI and the binned fit — the stat-only figure was an underestimate. *Relative to the binned fit with the same NPs*, however, there is a clean result: binning a (multivariate) discriminant is a deterministic *coarsening* of the per-event likelihood, so by the data-processing inequality it cannot carry *more* information, and profiling nuisances preserves the ordering (the profiled POI information is a Schur complement, which is operator-monotone). Hence **best-case NSBI CI $\leq$ binned CI, systematics included** — properly handling a systematic can only *shrink NSBI's advantage toward parity*, never reverse it (a reversal requires a modelling/coverage failure, excluded here).

Mechanism of the shrinkage. Profiling a systematic costs POI information in proportion to how much the systematic's effect *aligns with the POI's* in the space actually used. NSBI's gain lives in the feature dimensions beyond m_{HH}; a systematic that also lives there *and* points like κ_λ gives part of that gain back, while one confined to m_{HH} (already captured by binning) or orthogonal to the extra features does not.

Two caveats. (i) **Item 6 is the one genuine width give-back** even with perfect coverage: the morphing is a *learned* object estimated from the (smallest) $\pm1\sigma$ samples, so it carries a real estimation variance that a re-run-the-DNN template partly avoids — sample-starved $\pm1\sigma$ sets are where NSBI can pay *more* width (mitigate with more variation MC or an information-pooling GP morphing). (ii) The *realistic* binned baseline is a binned *DNN score*, not 1-D m_{HH}: the proven $\sim$2.5$\times$ gain was measured against binned m_{HH}, so NSBI's margin over a good binned-DNN is smaller, and the part that survives unconditionally is the pure *unbinning* increment (no information lost to binning the score).

Challenge	Binned (best case)	NSBI (best case)	Net effect on width advantage
1 κ_λ–syst. m_{HH} degeneracy	profiling JES/τES eats κ_λ info (acute in m_{HH}; a binned-DNN recovers some)	extra features *break* the degeneracy ($\sim$12.8$\times\to$1.3$\times$ inflation)	**largest win**
2 $\sim$40 experimental NPs	each NP profiled per bin; width grows with count	same NPs, per-event; grows similarly	preserved (equal cost)
3 data-driven bkg systematics	per-bin nuisance on the template	full per-event S/B *if* high-dim ABCDnn + per-event morphing	preserved best-case; *eroded* if the DD bkg must be binned
4 κ_λ-coupled theory	κ_λ-dependent lnN/shape; widens at the σ-minimum	shared — a signal-*prediction* band, not reduced by per-event power	$\approx$ neutral (shared floor)
5 factorisation cross-term	re-run the DNN with both NPs shifted	non-factorised (quadratic) morphing	preserved (both handle it)
6 MC-stat of learned ratios	Barlow–Beeston template MC-stat	estimation variance of the *learned* morphing (finite $\pm1\sigma$)	**genuine give-back**
a shower via CARL/DCTR	conservative 2-point *envelope*	profiled, data-constrained learned NP	**win** (narrows the envelope)
b absorb b-tag/τ-ID SFs	external POG prior	in-situ standard-candle constraint	**win** (in-situ > prior)

Where the width gain survives

Best-case, NSBI is never *wider* than binned; the advantage is just non-uniform. **Biggest wins:** #1 (degeneracy broken by dimensionality) and the upsides (a,b) (conservative *envelopes/priors* become *in-likelihood* constraints). **Shared / neutral:** #4 (irreducible signal-theory cost), #2, #5 (same NP cost, both pay it). **The one give-back:** #6 (finite-sample MC-stat of the learned morphing). So the headline gain is most robust where it comes from a *real* information advantage (unbinning + degeneracy-breaking + in-fit constraint of over-conservative systematics), and it erodes only where the systematic is irreducible-to-both (#4) or must be *learned* from too-small samples (#6).

6 Statistics and inference

- **Data signal yield is tiny and intrinsic.** $\sigma \times \mathcal{B} \sim 2.5$ fb $\Rightarrow \mathcal{O}(10\text{–}100)$ selected signal events (Run-2 to HL-LHC), under large backgrounds — vs the off-shell tail at $\sim$15% of σ. This is unchanged by MC and keeps the measurement systematics-comparable. (It is also the regime where NSBI's per-event information helps *most*.)
- **MC/training stats are ample** ($\approx$500k–750k per κ_λ point), so the morphing/ratio NNs are well-trained — the "under-determined morphing" worry applies to the *systematic-variation* samples, not the nominal.
- **Validation & coverage.** CMS inherits decades-validated COMBINE (asymptotics + toys + Barlow–Beeston). NSBI needs a custom JAX/IMINUIT fit with an explicit Neyman con-

struction and ratio-quality diagnostics (C2ST-like) built and validated from scratch [ATLAS p. 64,89].

7 Synthesis: could the NSBI κ_λ result be *worse* than the CMS fit?

Reading what CMS *actually* does removes a strawman: CMS does **not** bin m_{HH}; it bins a **multivariate DNN score** (shape systematics propagated by re-running the DNN on varied inputs, [CMS p. 93]). So CMS already carries the same multi-dimensional systematic information NSBI does — **NSBI is not "exposed to more systematics" than the real CMS baseline.** Against CMS-as-done, the comparison reduces to:

- **What NSBI buys:** *unbinning* (no information lost to binning the score) and a κ_λ-*optimal* per-event ratio (CMS's DNN is one fixed discriminant, not optimal at every κ_λ). These are the off-shell ∼2× / FAIR-Universe ∼20% gains.
- **What NSBI costs:** a *learned* morphing layer (NSBI must *train* the per-event systematic weight; CMS *counts* its re-run-DNN template, exact up to MC stat) and the *custom coverage validation*. The danger of "worse than CMS" is the *learned layer being mis-calibrated / under-validated* — producing a tighter-but-mis-covering interval — not a heavier systematics burden.
- **The genuinely new $b\bar{b}\tau\tau$ problem is the data-driven backgrounds** (Sec. 4): if QCD/-fakes cannot be given a faithful per-event density, the unbinned NSBI may have to bin them anyway, partially eroding its advantage in exactly the most sensitive channel.

Protections (so the worst case is "≈ CMS," not "worse and unknown"): honest morphing uncertainty (GP-style); explicit coverage validation; and **binning the NSBI discriminant as a built-in cross-check** (the binned fit is a coarsening of NSBI, so a regression is detectable and you fall back).

8 Summary of new challenges

Bottom line

The NSBI *method* ports cleanly from off-shell to κ_λ. The new challenges are in the *inputs*, and rank roughly: (1) **data-driven backgrounds** (QCD, jet→ τ fakes) with no per-event density — the most awkward fit with NSBI, but *solved-with-caveats* by flow-based estimation (**ABCDnn**, Sec. 4); (2) the κ_λ–**systematic shape degeneracy** (JES/τES on m_{HH}) plus a much larger experimental-systematics surface, which makes *morphing accuracy + honesty* the limiting factor; (3) the κ_λ**-dependent theory** and **factorisation cross-terms** (non-factorisable morphing); (4) **irreducible single-Higgs**. None are NSBI-specific *except* the data-driven-background mismatch and the learned-morphing layer; the rest are equally CMS's problems, handled with the same NPs. Two scope caveats: NSBI's gain is in the κ_λ/k_{2V} *parameter constraint* (shape directions), *not* the SM-HH observation significance (rate-saturated, luminosity-set); and the genuine per-event edge is on *shape/correlated-operator* directions, not the rate-broken $\kappa_t \leftrightarrow \kappa_\lambda$ / $k_V \leftrightarrow k_{2V}$ ones.

	ATLAS off-shell 4ℓ	κ_λ in HH$\to b\bar{b}\tau\tau$ (new challenge)
POI	$\mu_{\text{off-shell}}$ (1 POI)	κ_λ ($+k_{2V}, k_V$, EFT); shape in m_{HH}
Final state	4 leptons, fully reco, clean	b-jets+τ_h+MET; SVfit; 3 channels
Backgrounds	one MC $q\bar{q}ZZ$ (interfering), CR-constrained	QCD (ABCD), DY (18-CR SFs), $t\bar{t}$ (CR-SF), **jet$\to$ τ fakes**, single-H, diboson
Data-driven bkg in NSBI	none needed	**QCD & fakes have no per-event density**
Experimental systematics	few, small (leptons)	many, large (JES×11, τES, b-tag×5, DeepTau, ...)
Syst. vs signal shape	largely separable	**JES/τES deform m_{HH} (degenerate with κ_λ)**
Theory–POI coupling	none	**κ_λ-dependent theory band (Run-3: $+6/-23\%$ at SM; sign-flips with κ_λ)**
Factorisation of NPs	safe	JES×τES cross-term in m_{HH}
Signal yield	∼15% of σ	$\mathcal{O}$(10–100) events (fb-scale)
MC training stats	—	ample (≈500–750k/κ_λ); watch $\pm 1\sigma$ samples
Stats framework	custom (built)	must port the custom fit; CMS has COMBINE

Glossary of abbreviations & jargon

Every acronym and piece of jargon used in this note, in plain language. Symbols/couplings first, then abbreviations alphabetically.

Symbols & couplings

κ_λ (κ_λ) — Higgs trilinear self-coupling strength relative to the SM (= 1 in the SM); the parameter of interest. Enters gg→HH through the *triangle* diagram and interferes with the *box*.

k_{2V} — strength of the quartic $VVHH$ contact coupling (= 1 in SM); accessed in the high-m_{HH} VBF tail.

k_V — single Higgs–vector-boson coupling strength (VVH, = 1 in SM).

κ_t — top-quark Yukawa coupling strength (= 1 in SM); enters HH via the box and triangle.

m_{HH} — invariant mass of the di-Higgs system; the observable carrying the κ_λ interference shape.

$\cos\theta^*$ — production angle of the Higgs in the HH rest frame; the second variable in CMS's 2D κ_λ reweighting.

H_T — scalar sum of jet transverse momenta.

M_{jj} — invariant mass of the two VBF "tagging" jets.

N_j, N_b — number of jets / number of b-tagged jets (the ABCD(nn) conditioning variables).

p_{T} — transverse momentum (momentum in the plane transverse to the beam).

P_{ref} — reference hypothesis: the single fixed model all NSBI classifiers estimate density ratios *relative to*.

$r(x)$ — per-event likelihood (density) ratio — the quantity NSBI learns and the fit consumes.

ρ — (Pearson/Spearman) correlation coefficient; here between κ_λ and a systematic.

σ — standard deviation / Gaussian width ("$\pm 1\sigma$" = a one-standard-deviation systematic variation); also denotes a cross-section.

triangle / box — the two gg→HH amplitudes: the κ_λ-sensitive *triangle* (via an s-channel Higgs) and the κ_λ-independent *box*; their interference shapes m_{HH}.

GeV, TeV — giga-/tera-electronvolt (energy/mass units).

Abbreviations & jargon

ABCD — data-driven background estimate from four regions defined by two (assumed-independent) variables; the signal region $D = (B \times C)/A$.

ABCDnn — a normalizing-flow generalization of ABCD that morphs simulated background into the data-driven background *shape*, conditioned on extra variables.

AN — (CMS) Analysis Note: internal documentation (e.g. AN-18-121).

anchors — the fixed reference points (a κ_λ grid, or the $\pm 1\sigma$ systematic variations) between which a smooth morphing is interpolated.

Asimov dataset — a single artificial "data = expectation" dataset used to compute *expected* sensitivity without generating toys.

ATLAS, CMS — the two general-purpose LHC experiments.

AUC — area under the ROC curve; a classifier separation metric (0.5 = no separation).

Barlow–Beeston — standard treatment of finite-MC-statistics uncertainty via per-bin nuisance parameters.

BDT — boosted decision tree.

binned / unbinned — a fit that histograms an observable into template bins vs. one that uses each event's per-event likelihood directly (no binning). The CMS baseline is binned; NSBI is unbinned.

bootstrap — resampling with replacement to estimate a statistical uncertainty (e.g. of a trained network).

CARL / DCTR — the classifier-reweighting family: train a classifier to learn a per-event reweighting between two models. CARL = Calibrated Approximate Ratio of Likelihoods; DCTR = Deep Classifier-based ReweighTing.

closure (test) — a check that a method reproduces a known truth (e.g. predicted = true background in a validation region).

C2ST — Classifier Two-Sample Test: train a classifier to distinguish two samples; AUC $\approx$ 0.5 means they are indistinguishable (good closure).

Combine — CMS's standard binned maximum-likelihood fitting framework (HistFactory-style; asymptotic formulae plus toys, with Barlow–Beeston MC-stat handling).

CR / SR / VR — Control / Signal / Validation Region.

decay mode — the specific hadronic τ decay channel (e.g. 1-prong, 3-prong, with/without π^0); τES has per-decay-mode components.

DeepTau — CMS deep-learning hadronic-τ identification discriminant.

Delphes — a fast, parametrised detector simulation used for quick studies before full simulation.

DNN — deep neural network.

DY — Drell–Yan ($Z/\gamma^* \to \ell\ell$) background.

EFT / SMEFT — (Standard-Model) Effective Field Theory: parametrize new physics as higher-dimensional operators.

ES — energy scale (as in JES, τES).

EW — electroweak.

ensemble (spread) — training several networks (different seeds/folds); their spread estimates the MC-statistical uncertainty, propagated as one nuisance.

ExtendedABCD — ABCD augmented with extra sidebands to correct residual correlation between the two defining variables (a more accurate yield).

extrapolation (CR$\to$SR) — applying a transfer/calibration learned in a control region to the signal region; its validity is the key data-driven-background caveat.

fake-factor — data-driven estimate of jet$\to \tau_h$ misidentification: a transfer factor measured in a τ-anti-isolated CR and applied to the SR.

Fisher information — quantifies how sharply an observable constrains a parameter; here used to show that added dimensions shrink the κ_λ–systematic degeneracy.

flavour-tagging — identifying a jet's originating quark (b/c/light) with an ML discriminant (b-tagging is the b-jet case).

F-score — harmonic mean of precision and recall; here a separation proxy.

ggF / VBF — gluon–gluon fusion / vector-boson fusion (the two HH production modes).

ggH, VBF H, VH, $t\bar{t}H$ (single Higgs) — the single-Higgs production modes (gluon fusion / vector-boson fusion / associated with a vector boson / with a top pair); the irreducible same-final-state background (distinct from ggF/VBF for HH).

GP — Gaussian Process: a non-parametric model giving predictions with calibrated uncertainty bands.

HH — di-Higgs (Higgs-pair) production; **HHVV** = the quartic $VVHH$ contact vertex (set by k_{2V}).

Hesse / Hessian — matrix of second derivatives of the NLL; its inverse gives parameter covariances (MINUIT's HESSE).

HL-LHC — High-Luminosity LHC.

ID — identification (e.g. τ-ID, b-ID).

iminuit — Python interface to the MINUIT minimizer used for the profile fit.

interference — the quantum cross-term between two amplitudes (here triangle$\times$box in HH); it carries the κ_λ shape information.

ISR / FSR — initial-/final-state radiation (a parton-shower systematic).

JAX — an autodiff / accelerated-array library used to build the differentiable likelihood.

JER / JES — jet energy resolution / jet energy scale (systematics).

KL divergence — Kullback–Leibler divergence: a measure of difference between two probability distributions.

KS test — Kolmogorov–Smirnov goodness-of-fit test (a

p-value for "same distribution").

LHC — Large Hadron Collider.

lnN — log-normal: the usual constraint shape for a *normalization* nuisance parameter.

LO / NLO — leading / next-to-leading order in perturbative QCD.

MAE — mean absolute error.

MC — Monte Carlo (simulated events).

ME — matrix element.

MET — missing transverse energy (momentum imbalance, here mainly from neutrinos).

ML — machine learning.

mixture model — writing the total per-event density as a weighted sum of signal/interference/background components (the NSBI fit model).

MMD — Maximum Mean Discrepancy: a distribution-matching loss (used to train ABCDnn).

morphing — smoothly deforming a per-event density or weight as a function of a parameter (κ_λ or a nuisance), built by interpolating between anchors.

NAF — Neural Autoregressive Flow: a flexible normalizing-flow density model (ABCDnn's network).

Neyman construction — the frequentist procedure for building confidence intervals with correct coverage.

NLL — negative log-likelihood.

NN — neural network.

NP — nuisance parameter: a non-POI (systematic) parameter that is profiled in the fit.

NSBI / SBI — (Neural) Simulation-Based Inference: inference from learned per-event likelihood ratios instead of binned templates.

PAS — (CMS) Physics Analysis Summary, a short public document.

PDF — parton distribution function (separately, also "probability density function").

Perlmutter — the NERSC GPU supercomputer used for training.

pivoting / decorrelation — training a classifier to be *insensitive* to a nuisance (statistically independent of it).

POG — (CMS) Physics Object Group; provides standard object calibrations / scale factors.

POI — parameter of interest (here κ_λ).

poly-exp — polynomial×exponential interpolation used between systematic ($\pm 1\sigma$) anchors.

Powheg — an NLO matrix-element event generator.

profile (likelihood) — maximizing the likelihood over the nuisance parameters at each value of the POI.

PS / PU — parton shower / pile-up (additional simultaneous pp collisions).

Pythia, Herwig — two parton-shower/hadronization generators; their difference gauges shower-modeling systematics.

QCD — quantum chromodynamics; "QCD multijet" = the data-driven multijet background.

RealNVP — a specific (identity-initializable) normalizing-flow architecture.

resolution — the smearing/precision of a measured quantity (cf. JER for jets); "resolution handles" = input features encoding it.

RMS — root-mean-square.

RS3L — Re-Simulation-based Self-Supervised Learning (a representation-robustness method).

Run-2 / Run-3 — LHC data-taking periods (Run-2: 2016–18; Run-3: 2022–25).

SF — scale factor: a data/MC correction applied to reconstructed objects (b-tag, τ-ID, ...).

sideband — a region adjacent to (but outside) the signal region, used to estimate backgrounds; the extra regions in ExtendedABCD are sidebands.

SM — Standard Model.

Sophon — the (Sophon-AK4) jet "foundation model" backbone fine-tuned in the ML-pipeline studies.

SVfit — algorithm that reconstructs the $\tau\tau$ invariant mass from the visible decay products + MET.

THU$_{HH}$ — theory uncertainty on the HH cross-section/shape; κ_λ-dependent (Run-3 ggF scale+m_t: $+6/-23\%$ at the SM; the m_t-scheme asymmetry sign-flips across the κ_λ scan — AN-25-103 Table 40).

τES (tauES) — τ energy scale (a systematic).

toy / toy MC — synthetic pseudo-datasets drawn from the model, used for coverage and sensitivity tests.

trigger — the online selection deciding which collision events to record; carries its own efficiency scale factor.

VH / VZ — associated production of a Higgs / Z with a vector boson.

VIF — variance inflation factor: how much a parameter's variance grows due to correlation with others (a degeneracy metric).

WG — working group (e.g. the LHC HH WG).

ZZ, 4ℓ — a Z-boson pair / four charged leptons (the ATLAS off-shell final state).

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[4] A. Bahl, T. Plehn, N. Schmal, *global SMEFT fit of four di-boson processes with per-event likelihoods*, arXiv:2509.05409 (2025) — the $\sim$2$\times$ per-event gain on correlated-operator directions.

[5] ATLAS+CMS HH combination, arXiv:2602.23991 / CMS-PAS-HIG-25-014 (early 2026): $k_{2V} \in [0.73, 1.3]$, $\kappa_\lambda \in [-0.71, 6.1]$, $\mu_{HH} < 2.5$ (95% CL).

[6] A. Ghosh, M. Klute, J. Pan, *Focused Neural Simulation-Based Inference for the Higgs Self-Coupling in the $b\bar{b}\tau\tau$ Channel*, working note (22 Jun 2026); code: `NSBI-pheno/dihiggs_bbtautau`. Signal $gg \to HH$ from POWHEG at NLO with $\kappa_\lambda \in \{0, 1, 5\}$ (three-point quadratic morphing basis); reference $\kappa_\lambda = 5$; ten event-level observables; full background model with weighted fake-τ; unbinned extended likelihood vs binned m_{HH}, plus a focused test statistic. Detector response: **CMS full simulation** for the current validation, with a DELPHES fast-sim reproduction planned for the published phenomenological version (the setup the note describes). Results: κ_λ= 0 rejection 2.75σ (450 fb^{-1}) $\to$ 7.1σ (HL-LHC); 68% interval $\sim$10$\times$ tighter than the binned fit; 500-toy coverage validated. *No experimental-systematic nuisances in the fit.*

[7] R. Schöfbeck *et al.*, *Unbinned measurements with machine-learned systematic uncertainties* (GOLLUM): method MLST 6 015007 (2025), tested PRD 112 052006 (2025). Per-event parametric NP morphing to arbitrary order, factorizable *and* NP$\times$NP non-factorizable; tested on FAIR Universe ($\sim$3 calibration NPs, simulated single-lepton data). Treats NPs as POI-*independent*.

[8] P. Shyamsundar, *Comment on "An implementation of neural simulation-based inference for parameter estimation in ATLAS"*, arXiv:2505.19156 (2025). Argues, via a toy counterexample, that the ATLAS double-bootstrap estimator of the finite-simulation (MC-stat) uncertainty does not capture it — a correctness critique, not merely an under-validation. Dissection: the second bootstrap level resamples the ensemble *conditional on the fixed training set*, so it measures only the finite-ensemble noise ($\sim 1/M$, $\to 0$ as $M \to \infty$) and not the finite-sample error ($\propto 1/N$); Gaussian toy: true $\sim$0.1 vs reported $\sim$0.003 ($\sim$30$\times$). Proposed fixes: plain bootstrap without merging; fresh ensembles per pseudo-experiment ($M \times K$ retrainings); or summary-statistic SBI. Bears on item 6.

[9] S. Benevedes, J. Thaler, *Frequentist Uncertainties on Neural Density Ratios with wifi Ensembles*, arXiv:2506.00113 (2025). wifi = "$w_i f_i$": freeze M trained nets as basis functions; convex re-fit of the linear weights in log-ratio space on a held-out split (symmetrised MLC loss); covariance from the sandwich estimator $C = V^{-1}UV^{-1}$; per-event band $f_i C_{ij} f_j$; downstream fits inflated in closed form via the Gong–Samaniego theorem — no bootstrap, no extra NP. Nominal coverage on a Gaussian toy ($M \geq 8$) and q/g SBI ($M = 32$, Bootstrap protocol); *uniform-weight (naive) ensembles come out biased and under-covering in the same tests* (the second strike against spread-style estimators). Leading candidate for item 6; per-event band also feeds the honest morphing-uncertainty of item 9/#15. *Not addressed:* weighted/negative-weight events (our derivation task); untested on interference/multi-component mixtures.

[10] *Neural Fake Factor Estimation* (data-based density-ratio inference), arXiv:2511.06972, JHEP 04 (2026) 188. Per-event continuous fake factor as a learned density ratio (NSBI-compatible); *defers* fake-rate systematics to future work. Bears on items 3 and the data-driven backgrounds of Sec. 4.

[11] CMS Collaboration (A. Andrade *et al.*), *Search for non-resonant Higgs boson pair production in the $b\bar{b}\tau\tau$ final state with Run-3 data*, CMS AN-25-103 (draft, 2026); 172.3 fb^{-1} at 13.6 TeV + Run-2 combination. Key for this note: exact 3-component ggF signal morphing at fit time (Eq. 52–55; NLO samples at $\kappa_\lambda = 1, 2.45, 5$; VBF 6-component analog); theory uncertainties as *normalisation-only* lnN (Sec. 7.1, Table 42; ggF scale+$m_t = +6/-23\%$ at SM — the leading $b\bar{b}\tau\tau$ systematic); the band tabulated *per κ_λ point* with the m_t asymmetry sign-flipping (Table 40); pointwise scans and limit-vs-σ_{theory} exclusions (Secs. 8.4.5–8.4.6); joint QCD+m_t correlated across runs (Sec. 8.5.1). Basis of the C4a/C4b split in Sec. 5.